Enhancing Soil Salinity Mapping in Cropland by Integrating Agronomic Features With Remote Sensing

Caiyun Wen, Miao Lu, Yingqiang Song, Mingli Li, Yun Shi, and Wenbin Wu

***Abstract*—Soil salinization in cropland, particularly in arid and semiarid regions, is a major form of land degradation. Although remote sensing allows for soil salinity mapping by analyzing the spectral responses of soil and vegetation, a common limitation is the oversight of field management practices, which are known to significantly influence soil moisture dynamics and salinity accumulation. This study addresses this gap by integrating novel agronomic features—crop planting structure and irrigation indicator—into salinity prediction models. Crop planting structure was a quantitative representation of crop classification results using target encoding (TE). The irrigation indicator was derived from the temperature vegetation dryness index (TVDI) to capture seasonal soil moisture variations associated with irrigation during the growing season. Our results demonstrate that incorporating these features significantly improves model performance compared to conventional models using only environmental and spectral variables. The coefficient of determination (R^2) increased by 0.11, reaching 0.74. Notably, crop planting structure emerged as the most influential predictor, underscoring the value of field management information in cropland salinity mapping. Future research will explore advanced spatial representations of management practices, such as the distribution of irrigation infrastructure and continuous cropping years, to further enhance predictive reliability in soil salinity mapping.**

***Index Terms*—Crop planting structure, field management, irrigation, remote sensing, soil salinity mapping.**

I. Introduction

SOIL salinization is one of the most pervasive land degradation processes worldwide [1], [2], affecting over one billion hectares (≈10%–11% of global land) [3]. Irrigated agriculture is particularly threatened, with approximately 20% of irrigated land at risk [4]. This dynamic process, driven by both natural and anthropogenic factors [5], [6], reduces soil fertility and hinders crop productivity, posing a significant threat to sustainable agriculture and food security [7], [8], [9], [10]. Accurate mapping of soil salinity content (SSC) provides spatial distribution as basic data for informing suitable crop planting, soil management, and regional planning. However, traditional methods relying on dense soil sampling and laboratory analysis are prohibitively costly and time-consuming at regional scales.

Remote sensing provides a practical solution for monitoring soil salinity, offering broad spatial coverage and rich spectral information [11]. The reflectance of the soil surface, particularly during the bare soil period (the interval between crop harvest and sowing), is an important indicator of soil salinity [12]. During this period, salts accumulate at the soil surface, forming white salt crusts that distinguish saline soils from healthy ones. Increased SSC leads to higher soil reflectance in the visible-near-infrared spectrum [13], which can be detected using remote sensing. During the vegetation cover period, high soil salinity increases osmotic pressure, inducing physiological dehydration in crops, which can stunt growth or even cause death [14]. These salinity-induced crop health variations are detectable through remote sensing [15]. Therefore, remote sensing can capture both the direct spectral signature of salts and the indirect response of crops to salinity stress.

In recent years, many studies based on remote sensing imagery have been published on different scales. On a global scale, Wang et al. [16] quantitatively estimate global soil salt content at 10 m by integrating Sentinel-1/2 images, climate, parent material, terrain data, and machine learning. Wang et al. [17] combined topographic, meteorological, vegetative factors, and remote sensing indices to map soil salinity in space and time. On a regional scale, Zarif et al. [18] proposed an AI-driven soil salinity prediction approach for European soils using environmental covariates and remote sensing variables. Aksoy et al. [19] used 26 environmental covariates and spectral indices, two machine learning models, and two oversampling methods for soil salinity mapping. Within these studies, various remote sensing-derived features, including salinity indices (SIs) and vegetation indices (VIs), have been proven effective for soil salinity mapping [20], [21], [22], [23]. Compared to healthy cropland, saline cropland requires distinct field management practices, such as higher salt-tolerant crops and higher irrigation demands. Consequently, field management

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information represents a potent indicator for soil salinity mapping in cropland, yet it remains overlooked in related studies [24].

Agronomic features provide a more relevant indication of field management than spectral features. Farmers improve production in saline croplands through two approaches, including the adaptation of crops to saline soil [25] and the adaptation of saline soil to crops [26]. The integration of this activity information using remote sensing for soil salinity mapping would be an effective way to improve the inversion accuracy. The first approach of saline cropland improvement involves choosing planted crops based on soil salinity degrees [27], ecological constraints, and field infrastructure, guided by species-specific salt tolerance. Salt tolerance varies by species due to differences in ion toxicity and osmotic stress response [28]. For example, corn is salt-sensitive and is typically grown only in nonsaline or slightly saline fields, whereas sunflower has higher salt tolerance and is often planted in moderately saline areas [29]. Consequently, the observed crop types or planting structures serve as a reliable indicator of soil salinity. Recent work has confirmed that constructing separate salinity inversion models for different crop types improved prediction accuracy [30]. However, this approach is applicable to areas with a simple crop planting structure. In areas characterized by small-scale cultivation, constructing reliable soil salinity inversion models becomes challenging due to insufficient sample sizes for individual crop types.

The saline cropland improvement of saline soil adaptation to crops employs hydraulic leaching, drainage engineering, and soil improvements to reduce SSC in the tillage layer. Irrigation management plays a crucial role in controlling soil salinity in croplands. Farmers adjust irrigation timing based on salt accumulation, especially in arid and semiarid regions where rainfall is scarce. Consequently, fields with high salinity often exhibit relatively high soil moisture content during the crop growing season. Given this relationship, spectral indices representing soil moisture have great potential for soil salinity mapping in cropland. For example, the temperature vegetation dryness index (TVDI) integrates surface temperature and VIs to infer soil moisture conditions. Dong and Na [31] developed five salinity inversion models for different soil moisture levels, demonstrating that their approach achieved higher accuracy than traditional models. Therefore, integrating spatial variations in soil moisture into salinity inversion models can enhance prediction accuracy and improve large-scale soil salinity assessments.

In this study, we integrated two agronomic features for soil salinity mapping in cropland. The agronomic features included crop planting structure and irrigation indicator, which encode field management responses to salinity and, thus, provide information beyond environmental variables and spectral features. We hypothesized that combining agronomic features with conventional spectral and ancillary predictors will model soil–crop–water interactions and improve inversion accuracy. This study aimed to address the following points.

1) Extract agronomic features for soil salinity mapping and analyze the differences in SSC across various crop types and irrigation conditions.

2) Evaluate the effectiveness of integrating field management information by comparing the prediction performances of different feature combinations using environmental variables, spectral features, and agronomic features.

In summary, this article introduced a novel methodology that explicitly leverages field management information in soil salinity mapping, with the goal of achieving more accurate and reliable salinity maps in complex agricultural landscapes.

II. Study Area and Data

A. Study Area

The middle-upper reaches of the Yellow River, a major saline region in China, are affected by many factors such as geological structure, semidry climates, and rough agricultural irrigation. It makes soil salinization an environmental problem that affects agricultural production for a long time. Dalad Banner is located in the northern part of the region and in the western Inner Mongolia [see Fig. 1(a)] with a total area of 8241.07 km^2. The topography of the study area slopes from a higher elevation in the south, part of the Ordos Plateau, to a lower elevation in the north, within the Hetao Plain [see Fig. 1(b)]. According to the China Land Cover Dataset (CLCD) in 2022 [32], cropland in Dalad Banner is concentrated in the north, with barren (desert) in the centerwest and mainly grassland in the south [see Fig. 1(c)]. Agricultural statistics from the 2022 Dalad Banner Statistical Yearbook indicate that corn and sunflower are the major crops, accounting for 72.19% of the total sown area. The region exhibits a typical temperate continental monsoon climate with low precipitation and high evapotranspiration. Flood irrigation is the most common irrigation method, resulting in groundwater levels raising. Both climatic conditions and irrigation methods accelerate the accumulation of soil salts, leading to widespread saline areas.

B. Ground Samples

A total of 217 soil samples were collected in October 2022 [see Fig. 1(d)]. Each sample was obtained from a 10×10 m quadrat, aligning with the 10-m spatial resolution of Sentinel-2 imagery. Soil was sampled at a depth of 0–20 cm using a five-point sampling strategy. The geographic coordinates of each point were recorded using a handheld global positioning system (GPS) with horizontal positioning error <3 m. In the laboratory, the soil samples were air-dried, ground, and sieved through a 2-mm mesh. The SSC (g/kg) was determined by the gravimetric method using a 1:5 soil-to-water ratio. The soil samples were divided into five salinity degrees according to the saline soil classification standard of China [33]: nonsaline (SSC <1 g/kg), slightly saline (1–2 g/kg), moderately saline (2–4 g/kg), severely saline (4–6 g/kg), and saline soil (SSC >6 g/kg).

Ground crop samples are also necessary to obtain information on the spatial distribution of crops in the study area. A total of 352 crop samples, including corn, sunflower, rice, and other crops, were collected through a field survey in August 2022 during the crop growing season. During the sampling process, it was found that topsoil salinity characteristics (e.g.,

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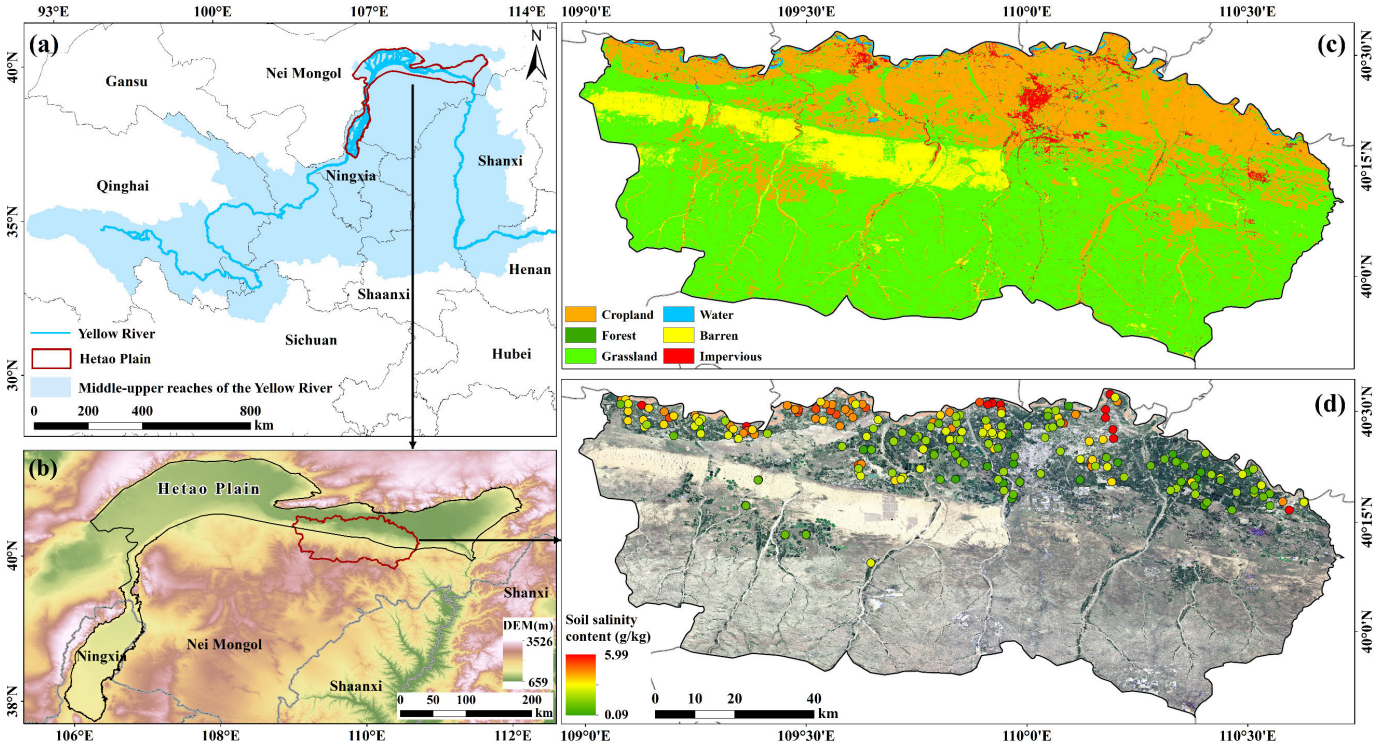


Fig. 1. Location of study area and soil samples. (a) Location of Dalad Banner in Inner Mongolia, China. (b) Topographic features of the study area. (c) Land cover types in 2022. (d) Spatial distribution of the soil sampling.

salt crusts) were more obvious in cropland planted with sunflowers than in cropland planted with corn. It is consistent with the fact that sunflowers have a higher salt tolerance than corn. Although rice is not highly salt-tolerant, the surface salinity is diluted because of the high-water demand during growth. Other crops include wheat, vegetables, and soybeans, with small planting areas and similar low salt tolerance.

C. Sentinel-2 Imagery

The Sentinel-2 is one of the most commonly used multispectral imagers in remote sensing. It has been proven that Sentinel-2 performed better than Landsat for soil salinity prediction [20]. The Sentinel-2 consists of two high-resolution multispectral imaging satellites (Sentinel-2A and Sentinel-2B), offering a time resolution of five days at the equator. The imagery has 13 bands with a spectral range of 400–2400 nm, covering visible, near-infrared, and short-wave infrared, and the spatial resolution is 10 m in visible, 20 m in near-infrared, and 60 m in short-wave infrared, respectively. According to the phenological calendar and Google Earth Engine (GEE) platform, six cloud-free images from June through October 2022 (2022-06-07, 07-07, 08-11, 08-31, 09-25, and 10-10) were collected for crop classification. The smallest interval between sampling date and satellite data acquisition date is also important to ensure the inversion accuracy [20]. Therefore, the October 2022 image was obtained for soil salinity inversion.

D. Environmental Variables

The environmental variables include a topographic feature and soil properties. The topographic feature was derived from the ALOS PALSAR digital elevation model (DEM) with a spatial resolution of 12.5 m, which is the highest-resolution DEM publicly available. Soil properties were sourced from national gridded maps of soil properties at 90-m resolution, covering six standard depth intervals across China (0–5, 5–15, 15–30, 30–60, and 60–100 cm) [34]. The soil properties in the 0–20-cm depth were calculated by a depth-weighted average function using three depth intervals (0–5, 5–15, and 15–30 cm) to be consistent with the depth of soil sampling [35], [36]. Two soil properties closely related to SSC, cation exchange capacity and pH, were chosen for the soil salinity inversion. All data were resampled to 10 m for the consistency of the spatial resolution.

III. Methodology

The methodology was structured into three steps, including crop planting structure extraction, irrigation indicator calculation, and soil salinity mapping (see Fig. 2). Detailed descriptions of each step were provided in Sections III-A–III-C.

A. Crop Planting Structure Extraction

1) Crop Classification: The accurate spatial distribution of major crops is the basis for extracting crop planting structure. The crop distribution map is generated through a process of crop classification by using time-series images and ground samples. First, crop type labels from ground samples are correlated with the corresponding multitemporal spectral profiles from satellite imagery. Then, a classifier learns this relationship to build a classification model and is finally applied to all pixels in the study area, thereby converting discrete point

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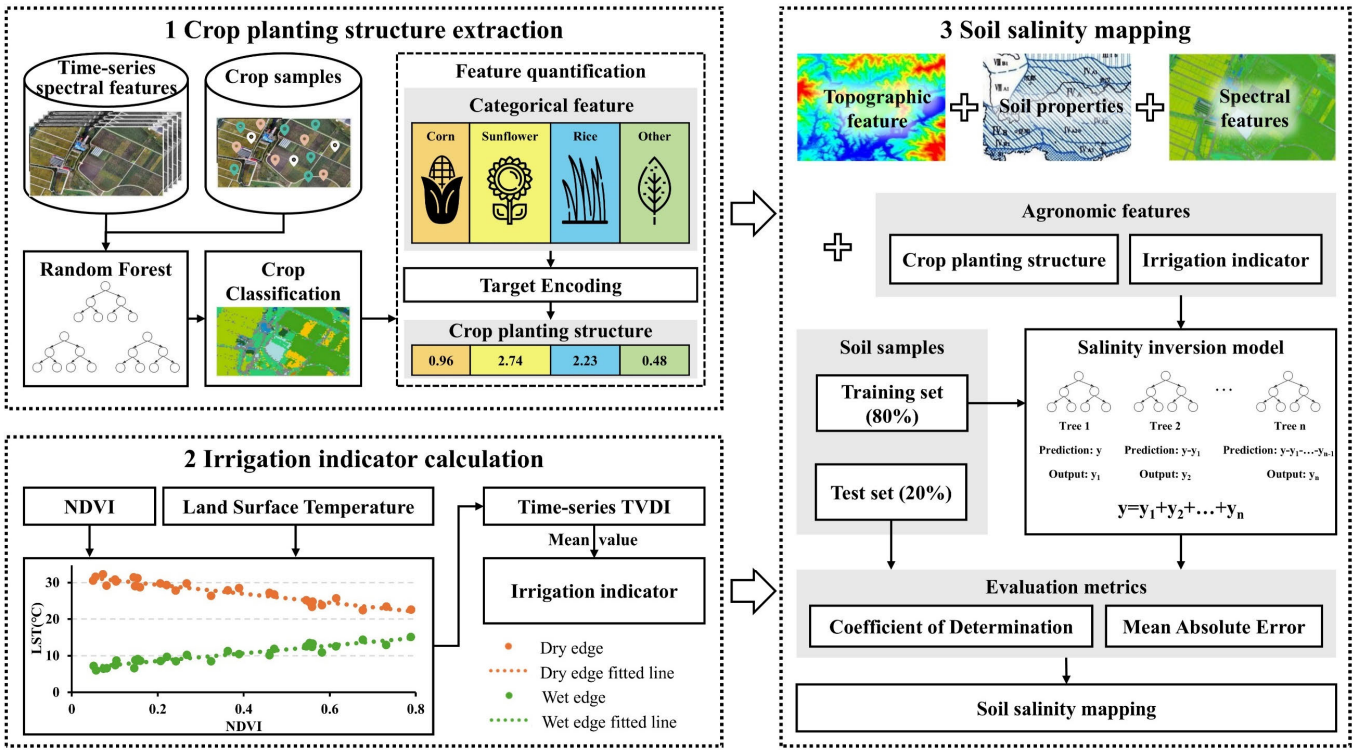


Fig. 2. Research flowchart for soil salinity mapping in cropland.

data into a continuous spatial map. The combination of time-series spectral features and VIs with a random forest (RF) classifier has been demonstrated to obtain crop classification results with sufficiently high accuracy [37]. In this study, six images were acquired from June to October 2022 based on the crop phenological calendar for Dalad Banner. Six VIs were calculated, including the green vegetation index (VIgreen) [38], the normalized difference vegetation index (NDVI) [39], the normalized difference tillage index (NDTI) [40], the normalized difference senescent vegetation Index (NDSVI) [41], the land surface water index (LSWI) [42], and the enhanced vegetation index (EVI) [43]. The cropland distribution from the 2022 CLCD was extracted and masked to support subsequent classification. Target crop types included corn, sunflower, rice, and other crops, with other crops integrating wheat, vegetables, and soybean. The crop samples were randomly divided into training and test sets according to 4:1. The training set was used to construct the classifier, while the test set was used to evaluate the performance of the classifier. The classification results were assessed using the overall accuracy (OA) and the kappa coefficient.

2) Crop Planting Structure Quantification: Machine learning-based salinity inversion models required all input features to be numerical variables, so it was necessary to quantify the sole categorical variable, crop type, into a numerical variable. This transformation was achieved through target encoding (TE) [44], [45]. Each soil sample was able to obtain information about the corresponding crop type from the crop classification result. In the case of corn, the training set of soil samples is screened for samples that fall on the corn pixels, and then, the SSC mean of these samples is calculated. If the mean value is 1.50 g/kg, replace all the corn pixel values in the crop classification results with 1.50. The above steps process sunflower, rice, and other crop pixels sequentially to finally obtain the quantitative crop planting structure.

The TE approach offers distinct advantages over conventional encoding methods. On the one hand, it preserves the ecological significance of crop-salinity relationships by embedding observed SSC-crop associations directly into the feature values, enhancing the model interpretability. On the other hand, it generates numerical outputs that maintain compatibility with algorithmic processing while avoiding the dimensionality expansion issues inherent to one-hot encoding.

B. Irrigation Indicator Calculation

The irrigation indicator was proposed to characterize the irrigation practices of farmers, thereby reflecting the degree of soil salinity in cropland. The TVDI [46], which associates land surface temperature (LST) with NDVI, has been widely applied to assess vegetation and soil water stress [47]. The Landsat 9 images from June to October 2022 were acquired from the GEE platform. The images were preprocessed by radiometric calibration, atmospheric correction, and cloud removal. Then, the data were resampled to 10 m and were used to calculate the LST and NDVI. The time-series TVDIs were determined by using the following formulas [46]:

$$TVDI = \frac{T_s - T_{s\min}}{T_{s\max} - T_{s\min}} \tag{1}$$

$$T_{s\max} = a_1 NDVI + b_1 \tag{2}$$

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$$T_{s\min} = a_2 NDVI + b_2 \tag{3}$$

where T_s is the observed LST at the given pixel, NDVI is the normalized vegetation index value of the pixels, and $T_{s\max}$ and $T_{s\min}$ are the maximal and minimal LSTs of the dry and wet edges corresponding to the specified NDVI in the target region, respectively. $T_{s\max}$ and $T_{s\min}$ are linearly expressed using two statistical coefficients (a, b), as shown in (2) and (3).

The mean TVDI was calculated as the irrigation indicator using time-series TVDIs during the crop growing season. It effectively captures the spatial and temporal characteristics of agricultural water management. The low values of the irrigation indicator indicate intensive irrigation practices employed to leach accumulated salts from the tillage layer. Consequently, these lower values serve as a reliable diagnostic indicator of a high degree of soil salinization in cropland.

C. Soil Salinity Mapping

We developed an inversion model by integrating environmental variables, spectral features, and agronomic features for soil salinity mapping. The environmental variables included topographic features (DEM) and soil properties (cation exchange capacity, pH). The spectral features included SIs contemporaneous with soil sampling [48] (see Table I) and time-series NDVIs. The agronomic features included crop planting structure and irrigation indicator. To optimize feature space and enhance computational efficiency, we applied principal component analysis (PCA) to the SIs and time-series NDVI, respectively. The cumulative contribution rate is the criterion for judging the number of selected principal components (pc). Specifically, principal components achieving a cumulative variance explanation exceeding 80% were retained for model integration, ensuring optimal information preservation while maintaining dimensionality reduction benefits.

We employed the extreme gradient boosting (XGBoost) algorithm to develop our soil salinity inversion model. As an advanced ensemble learning method derived from gradient boosted decision trees (GBDTs), XGBoost enhances computational efficiency and generalizability [49]. XGBoost is an ensemble learning method that iteratively improves prediction accuracy by additively training new models to predict the residuals of previous models. The gradient boosting method employs gradient descent optimization to systematically minimize the loss function when integrating each new model into the ensemble. Based on salinity degrees, soil samples were randomly stratified at 4:1 for training and test sets. The training set was used to construct the inversion model, while the test set was used to evaluate the performance of the predicted results.

D. Comparison Experiments

We systematically evaluated the contribution of agronomic features to soil salinity mapping accuracy through six feature combinations (see Table II). The experimental design comprised two distinct feature types as follows.

1) S1–S3 assessed the baseline performance using commonly used features, where S1 incorporated fundamental environmental variables (topographic indices and soil properties), S2 augmented S1 with SIs, and S3 further integrated time-series NDVIs to evaluate the incremental value of multitemporal remote sensing data.

2) S4–S6 extended these baselines by progressively integrating agronomic features (crop planting structure and irrigation indicator) to quantify their additional predictive power. This hierarchical experimental design enabled the comparison of the impact of different feature types on improving the accuracy of soil salinity mapping in cropland.

The performance of the soil salinity inversion model was evaluated using two metrics: R-square [R^2; see (4)] and mean absolute error [MAE; see (5)]. A larger R^2 and a smaller MAE indicate superior model performance

$$R^2 = 1 - \frac{\sum_i (y_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2} \tag{4}$$

$$MAE = \frac{1}{m}\sum_{i=1}^{m} |(y_i - y_i)| \tag{5}$$

where y_i is the predicted value of the ith sample, y_i is the true value of the ith sample, $\bar{y}$ is the average SSC of soil samples, and m is the total number of soil samples.

IV. Results

A. Statistical Analysis of Soil Samples

The salinity content range of soil samples was 0.09–5.99 g/kg, with a mean of 1.35 g/kg and a standard deviation of 1.11. The statistical characteristics of SSC were analyzed based on crop planting structure [see Fig. 3(a)] and grouped irrigation indicator [see Fig. 3(b)]. According to the crop type corresponding to the soil sample, Fig. 3(a) illustrates the SSC distribution variations among crop types, including mean, standard error (SE), and median of soil salinity for different crop types. Sunflower samples spanned a broad salinity range from slightly to severely saline (1–6 g/kg). Rice was mainly distributed in slightly and moderately saline soils (1–4 g/kg). Corn and other crops were mainly grown in nonsaline and slightly saline soils (0–2 g/kg). Mean SSC was highest in sunflower, followed by rice, corn, and other crops, consistent with the order of crop salt tolerance.

The irrigation indicator ranging from 0 to 1 was divided into five levels: very wet (0.0< irrigation indicator ≤ 0.2), wet (0.2< irrigation indicator ≤ 0.4), water balanced (0.4< irrigation indicator ≤ 0.6), dry (0.6< irrigation indicator ≤ 0.8), and very dry (0.8< irrigation indicator ≤ 1.0) [50]. The soil samples were distributed in the first four classes [see Fig. 3(b)]. The salinity degrees in the very wet level were mainly slightly and moderately saline due to the excess water supply, where the corresponding crop type is mainly rice. In contrast, SSCs of samples in the wet, water-balanced, and dry levels exhibit progressively lower salinity degrees. The wet level indicates higher soil moisture content during the crop growing season. Since the study area is located in the arid and semiarid zone with low natural precipitation, the wet level suggests more frequent supplemental irrigation by farmers and higher soil salinity. The dry level indicates less irrigation and healthy crop growth under natural conditions.

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TABLE I
DESCRIPTION OF FIVE SIS USED FOR SOIL SALINITY INVERSION

Name	Calculation formula
SI_1	$\frac{NIR \times Red}{Green}$
SI_2	$\frac{Blue - Red}{Blue + Red}$
SI_3	$\sqrt{Red^2 + Green^2 + NIR^2}$
SI_4	$\sqrt{NIR \times Red}$
Canopy Response Salinity Index (CRSI)	$\sqrt{\frac{NIR \times Red - Green \times Blue}{NIR \times Red + Green \times Blue}}$

TABLE II
DIFFERENT FEATURE COMBINATIONS FOR SOIL SALINITY MAPPING

ID	Feature combination
S1	DEM + Soil properties
S2	DEM + Soil properties + SIs
S3	DEM + Soil properties + SIs + NDVIs
S4	DEM + Soil properties + SIs + NDVIs + Crop planting structure
S5	DEM + Soil properties + SIs + NDVIs + Irrigation indicator
S6	DEM + Soil properties + SIs + NDVIs + Crop planting structure + Irrigation indicator

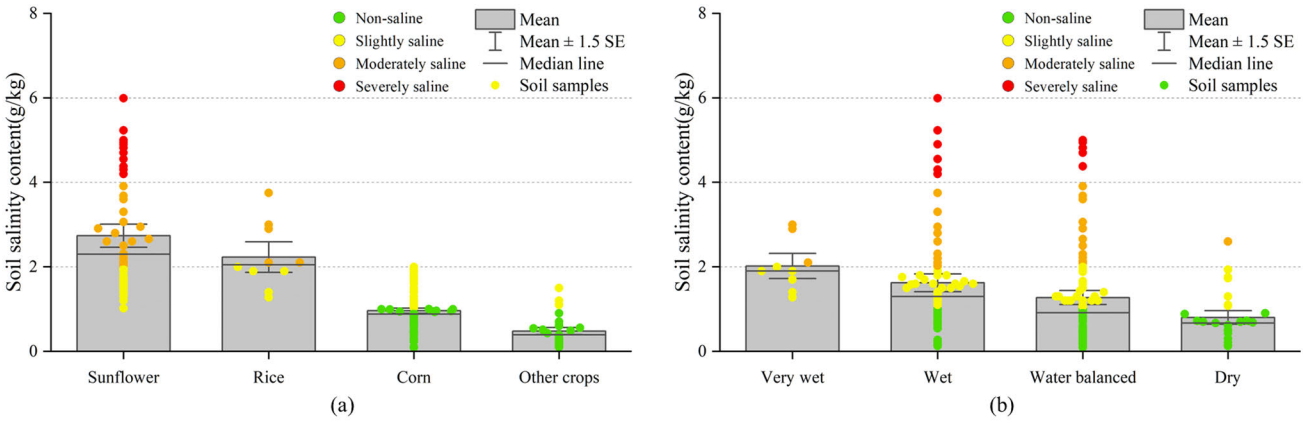


Fig. 3. Distribution of soil samples across different crop types (a) and different irrigation conditions (b) SE is calculated by dividing the sample standard deviation by the square root of the sample size.

B. Spatial Distribution of Crop Planting Structure

The results of the spatial distribution of different crops in Dalad Banner are presented in Fig. 4(a). The combination of time-series spectral features and VIs with the RF classifier obtained the OA of 97.17% and Kappa of 0.96. Corn is the most widely planted staple crop, mainly distributed in the northern region. Sunflowers are primarily concentrated in the western region. Rice planting areas are relatively small, mainly situated along the Yellow River irrigation zone in the northwestern region. Other crops are primarily located in the central and southwestern parts, where the predominant irrigation method is sprinkler irrigation, leading to the distribution of these crops in circular cropland fields.

Crop types were quantified as mean SSC using the TE method. The mean SSC of training samples and all soil samples is shown in Table III, proving that the SSC distribution in the training set closely aligns with that of all samples. The sunflower exhibited the highest mean SSC, followed by the

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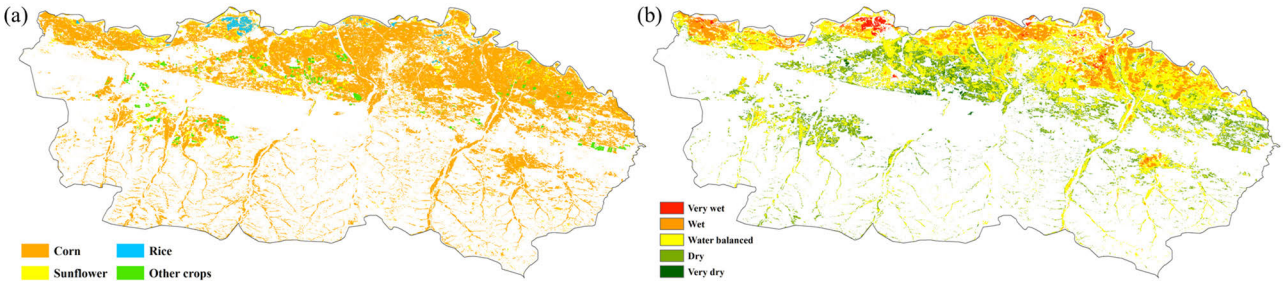


Fig. 4. Spatial distribution of crops (a) and the irrigation indicator (b) in Dalad Banner.

TABLE III
AVERAGE SSC OF TRAINING SAMPLES FOR DIFFERENT CROPS

Crop type	Mean SSC of training samples (g/kg)	Mean SSC of soil samples (g/kg)
Sunflower	2.74	2.74
Rice	2.28	2.25
Corn	0.95	0.96
Other crops	0.46	0.48

rice, corn, and other crops. Four mean values were substituted into the crop classification result to derive the crop planting structure, which was subsequently incorporated into the spatial prediction of soil salinity.

C. Spatial Distribution of Irrigation Indicator

The analysis of the irrigation indicator provides a comprehensive overview of the spatial distribution of the irrigation condition across the study area. In this study, the mean TVDI was calculated as the irrigation indicator over the crop growing season [see Fig. 4(b)]. Regions with lower irrigation indicator values were mainly located in the north, which is close to the Yellow River with a sufficient water supply. The irrigation conditions included very wet and wet. Rice was mainly grown in very wet areas. Regions with higher irrigation indicator values were located in the central and southern parts of Dalad Banner. The water-balanced region was the largest and most widely distributed. This spatial variability in the irrigation indicator indicates the influence of anthropogenic factors on irrigation practices, providing reference information for soil salinity mapping in cropland.

D. Spatial Distribution of Soil Salinity

In this study, we chose the crop planting structure and irrigation indicator as the specific characterization of field management while integrating topographic features, soil properties, and spectral indices for soil salinity mapping. Due to the high dimensionality of the SIs and time-series NDVIs, PCA was applied separately to each dataset to centralize valid information for feature dimensionality reduction. The contributions of the $pc1_{SI}$ and $pc2_{SI}$ were 57.26% and 37.92%. The contribution of the $pc1_{NDVI}$ was 83.64%. Therefore, three principal components were added to the salinity inversion model.

The predicted spatial distribution of soil salinity is shown in Fig. 5(a). R^2 of soil salinity mapping using environmental variables, spectral features, and agronomic features was 0.74, and the MAE was 0.36. The SSCs in the Dalad Banner showed a discernible trend of highest in the north, lower in the south, and lowest in the center. The SSCs along the Yellow River are all higher in the northern plain. The high SSC areas in the northwest are mainly planted with rice, and the high SSC areas in the northeast are mainly planted with sunflower, thus showing a clear aggregation pattern. It is more evident in the salinity degree map [see Fig. 5(b)]. Nonsaline cropland constituted the largest portion of the study area, covering 128 974 hectares (48.17%), followed by slightly saline areas at 123 622 hectares (46.17%). Moderately and severely saline lands accounted for significantly smaller areas, with 14 789 hectares (5.52%) and 344 hectares (0.13%), respectively.

E. Comparison of Feature Combinations

Six different feature combinations were input into the inversion model to assess the impact of agronomic features on the accuracy of soil salinity mapping, as shown in Fig. 6. The worst result was presented by S1, which only included the DEM and soil properties, with R^2 of 0.25 and MAE of 0.55 g/kg. By comparing S1 and S2 (DEM + soil properties + SIs), the addition of SIs increased R^2 by 0.29, indicating that remote sensing data can effectively improve the accuracy of soil salinity mapping at a large scale. The results of S2 and S3 (DEM + soil properties + SIs +VIs) revealed that time-series NDVIs were also useful for predicting soil salinity, where R^2 increased to 0.09. S3 was used as a baseline to verify the effectiveness of the agronomic features. Compared with the commonly used feature combination S3, all the other modeling scenarios integrating agronomic features enhanced

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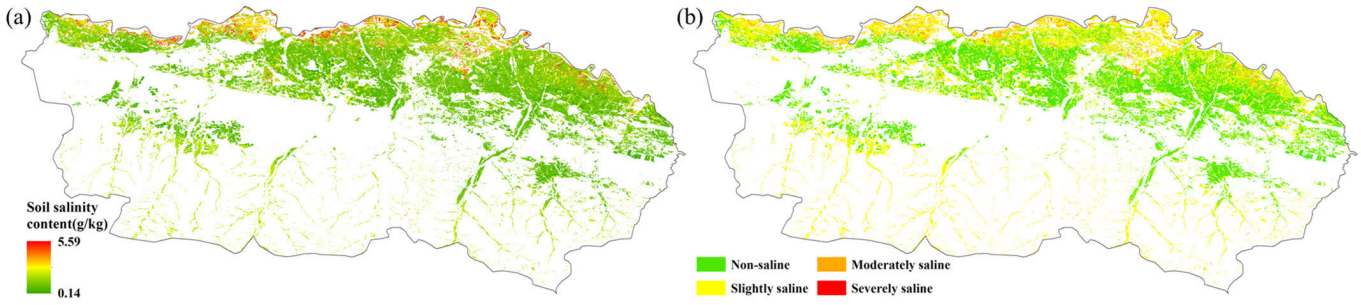


Fig. 5. Spatial distribution of SSC (a) and salinity degrees (b) in Dalad Banner.

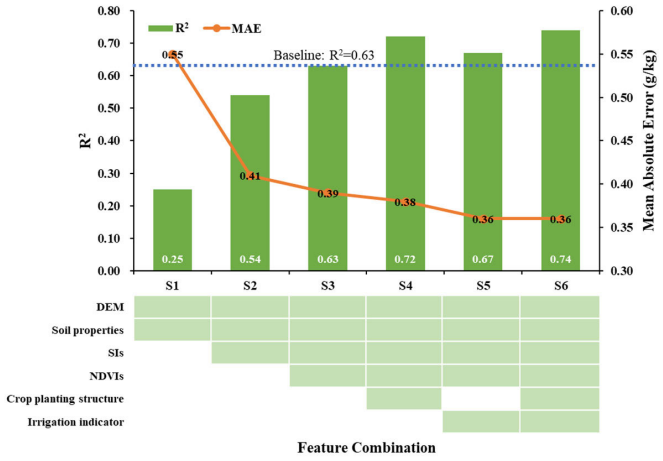


Fig. 6. Accuracy assessment of different feature combinations.

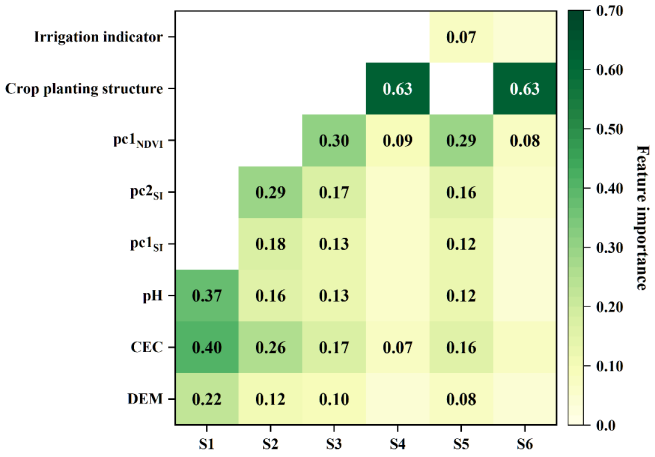


Fig. 7. Feature importance of different feature combinations. The feature importance of greater than 0.07 was labeled.

SSC predictability. S6 (DEM + soil properties + SIs + VIs + agronomic features) had a maximum R^2 of 0.74, followed by S4 (DEM + soil properties + SIs +VIs + crop planting structure) and then S5 (DEM + soil properties + SIs +VIs + irrigation indicator). The involvement of the crop planting structure significantly improved the inversion accuracy and generalization ability of the inversion model. The integration of the irrigation indicator reduced the MAE of the inversion model. The results showed that the inclusion of agronomic features was able to obtain a higher inversion accuracy than using the commonly used feature combination. It proved the effectiveness of agronomic features for soil salinity mapping, especially the crop planting structure.

The feature importance given by the XGBoost can evaluate the contribution of each variable to the inversion model using information gain (see Fig. 7). DEM was the least important of the inversion scenarios. Cropland in Dalad Banner is mainly located in the northern plain with little topographical variation. Two soil properties (i.e., CEC and pH) were influential variables in S1, S2, S3, and S5. Spectral indices (i.e., SI and NDVI) showed higher importance than environmental variables in S2, S3, and S5. The highest feature importance was the crop planting structure in S4 and S6. It indicates the effectiveness of crop planting structure for soil salinity mapping in cropland. Although the importance of the irrigation indicator is relatively low, R^2 increased by 0.04, and MAE decreased by 0.03 compared to S5 and S3. In summary, soil properties, spectral indices, and crop planting structure were important variables for estimating soil salinity in cropland.

V. Discussion

A. Spectral Feature Processing

We performed PCA to reduce the dimensionality of spectral features, including multiple SIs and time-series NDVI. Many studies demonstrated the efficacy of various SIs and VIs in improving inversion accuracy. However, the spectral indices are highly correlated due to overlapping spectral bands (e.g., visible and near-infrared) used in the linear calculation. To address this, PCA objectively condensed the information from all SIs and NDVIs into a few uncorrelated components, optimizing the inputs for the inversion model.

As shown in Fig. 8, the principal components obtained through PCA better reflect the differences among different salinity degrees compared with the original features. Fig. 8(a) shows the mean values of several SIs across four salinity degrees. In the raw form, these indices exhibit little separation between degrees, and most indices decrease as salinity increases (except SI2). The NDVI in July and August was separable between different salinity degrees [see Fig. 8(b)]. NDVI consistently decreased with higher salinity degrees, reflecting the suppressive effect of salt on crop growth. In contrast, the principal components of the SI and NDVI

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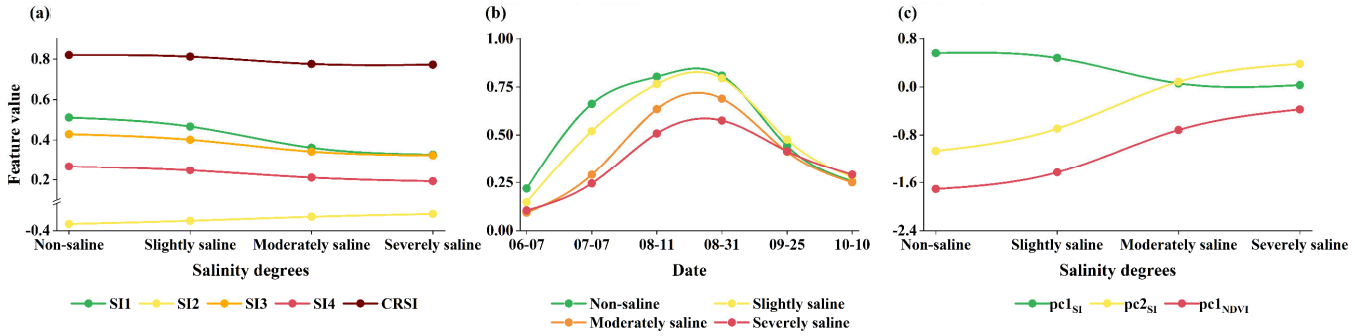


Fig. 8. Feature values of multiple (a) SIs, (b) time-series NDVI, and (c) principal components. Solid lines represent the mean value of samples.

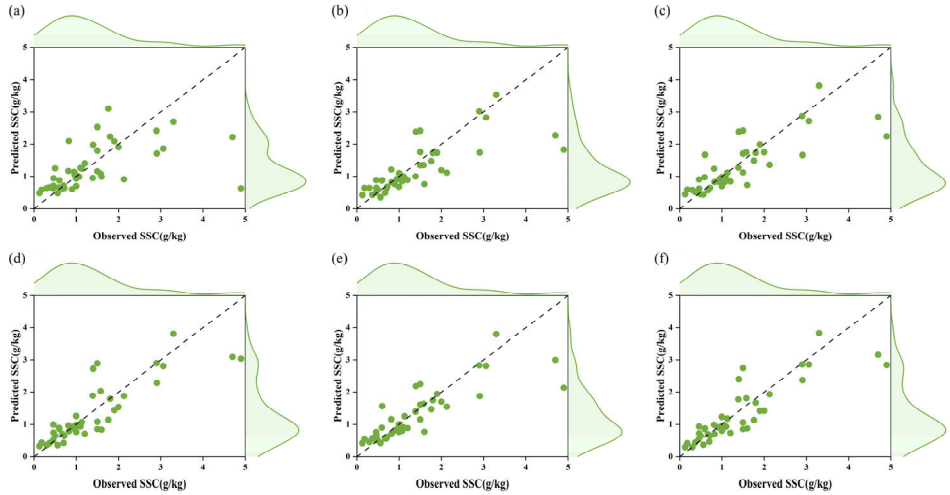


Fig. 9. Scatterplots of observed SSC versus predicted SSC for different feature combinations. The dashed line in the figure is the 1:1 line. The curves on the upper and right axes are the distribution curves of the observed and predicted values, respectively. (a) S1. (b) S2. (c) S3. (d) S4. (e) S5. (f) S6.

showed more pronounced separation of salinity degrees [see Fig. 8(c)]. The $pc1_{SI}$ retained the negative correlation of SIs with salinity, while $pc2_{SI}$ captured a complementary positive trend. This confirmed that the PCA-derived features amplified degree distinctions, providing the inversion model with more discriminative spectral inputs. Future studies will consider exploring nonlinear dimensionality reduction methods such as t-distributed stochastic neighbor embedding (t-SNE) or uniform manifold approximation and projection (UMAP) to further compare the feature extraction performance with PCA.

B. Predicted Results

The scatterplots of different feature combinations are shown in Fig. 9. The predicted results were compared with the observed SSC, predicted SSC, and the 1:1 line. The underestimation of SSC in the moderately and severely saline areas and the overestimation of SSC in the nonsaline and slightly saline areas were the basic cause affecting the inversion results. The distribution of predicted SSC in S1 differed significantly from that of observed SSC. The predicted values of S2, S3, and S5 had similar distribution features. R^2 of S4 and S6 with the integration of the crop planting structure was significantly higher, and it could be seen that the outliers with higher salinity were closer to the 1:1 line. When the observed SSC was greater than 4 g/kg, the points in Fig. 9 were distributed below the 1:1 line. This indicates that inversion models of S1–S6 underestimate SSCs of severely saline soils.

The underestimation of high soil salinity stems from two aspects, including the data distribution limitation and the inherent mechanisms of the XGBoost model. The proportion of slightly and moderately saline samples exceeds that of severely saline samples, consistent with the actual regional distribution. This imbalance in sample numbers prevents the model from adequately learning the characteristics of high-salinity samples, thereby compromising inversion accuracy. In addition, XGBoost is an ensemble of multiple decision trees. At each leaf node of the tree, the output represents a weight derived from all training samples within that node, optimized under a regularized objective function. This value approximates the mean residual of the contained samples. The regularization primarily serves to prevent overfitting, producing a smoother model with improved generalization. However, this smoothing effect systematically compresses the range of predicted values—reducing high predictions and elevating low ones—thereby concentrating outputs around the mean.

C. Advantages of Agronomic Features

This study is novel in integrating agronomic features with traditional environmental and spectral variables for a soil

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salinity inversion model. Based on current research, the inclusion of additional variables such as environmental variables and remote sensing indices does not significantly enhance the modeling accuracy of soil salinity inversion [11]. The two agronomic features we proposed were crop planting structure and irrigation indicator, both of which represent field management information. Our results have demonstrated that including these variables significantly improved traditional model performance.

Many methods have been developed to enhance the accuracy of soil salinity inversion in arid and semiarid regions. Wu et al. [51] effectively enhanced the sensitivity to salinity variations by transforming Landsat 8 reflectance using ground hyperspectral data, significantly improving salinity inversion accuracy in the Yinchuan Plain region with R^2 of 0.73. Wang et al. [52], in Southern Xinjiang, demonstrated that combining remote sensing with landscape characteristics using machine learning effectively enhances salinity mapping accuracy (R^2 = 0.75). Furthermore, Peng et al. [13] highlighted the robust stability and applicability of the Cubist model in soil salinity mapping across diverse land use types (R^2 = 0.91). However, these studies focused on salinity mapping across multiple land cover types, without concentrating on cropland—a typical land use of high anthropogenic disturbance. The higher spatial heterogeneity than other land use types results in greater modeling challenges [11]. For instance, Gao and Xu [53] achieved R^2 of 0.66 when mapping cropland soil salinity using a multicycle classification approach. Cui et al. [30] developed salinity inversion models for corn and sunflower in the Hetao Irrigation District, yielding R^2 of 0.72. In this study, integrating the crop planting structure and the irrigation indicator elevated R^2 to 0.74. This conclusively demonstrates the efficacy of integrating field management information for soil salinity inversion in cropland.

The crop planting structure provided the greatest improvement in accuracy for soil salinity mapping. Based on the differences in crop salt tolerance, farmers make planting decisions based on soil conditions. To some extent, crop planting structure is an amplification of relatively accurate information on soil salinity [54]. The basis of crop planting structure is the spatial distribution information of crop types. The crop information indicates the upper limit of salinity in cropland, thus narrowing the prediction range and improving the inversion accuracy. In this study, the high-accuracy crop classification result was obtained based on commonly used spectral bands and VIs during the crop growing season. Therefore, the acquisition of new features did not substantially increase the complexity of soil salinity mapping at a large scale while being able to improve the inversion accuracy. It is noted that the crop planting structure requires differences in salt tolerance of the major crops in the target area. For example, in the Yellow River Delta, most staple crops (wheat, soybean, and corn) shared similarly low salt tolerance, so the crop type information cannot provide an effective range of soil salinity distribution.

Our results also showed that the irrigation indicator, combined with commonly used features, also improved the inversion accuracy. Our field surveys revealed that farmers in severely saline regions adaptively adjust irrigation practices according to observed surface salt accumulation, thereby mitigating crop growth constraints caused by increased salinity contents [55]. The study area of this study is located in an arid and semiarid zone with low precipitation and high evaporation. Soil moisture comes mainly from irrigation rather than natural precipitation. Therefore, highly saline fields tend to be irrigated more intensively or frequently, which creates observable differences in our irrigation indicator. By integrating this feature, the inversion model could better distinguish salinity degrees based on these irrigation-driven patterns.

Future studies should further explore the substantial impacts of field management on saline soils by integrating additional agronomic features to enhance model accuracy. For example, continuous cropping years are one of the potential candidates, which may improve salinity inversion precision by capturing the relationship between continuous cropping and salinity degrees. The integration of such management-related variables with remote sensing could provide a more comprehensive understanding of agricultural salinization processes.

VI. Conclusion

To overcome the limitations of large-scale soil salinity mapping in cropland, we developed a novel salinity inversion model integrating environmental variables, spectral features, and agronomic features. The agronomic features included crop planting structure and irrigation indicator, which serve as proxies for field management practices. By comparing the accuracy with commonly used feature combinations, the results showed that the integration of agronomic features significantly improved inversion accuracy (R^2 increase of 0.04–0.11 from 0.63). The crop planting structure demonstrated greater predictive power than the irrigation indicator due to its direct integration of crop-specific salt tolerance mechanisms.

The agronomic features showed great potential for application in salinity mapping in cropland. They are particularly valuable in two scenarios, including regions with pronounced divergence in crop salt tolerance within local planting structures and regions with irrigated cropland rather than rain-fed cropland. These findings highlight the critical role of field management in the soil salinization process. Such advancements could further bridge the gap between remote sensing-based monitoring and on-ground salinity management practices, particularly in intensively cultivated regions facing progressive soil degradation. This framework could be further integrated into regional decision support systems and land management platforms to assist in monitoring and mitigating soil salinization.

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